Fluke 8840A | Diagnostic Error Codes

Bench reference • Complete documented error-code list from Instruction Manual, Table 2-1

Analog self-test errors | 01–21

Code	Meaning / test configuration
01	200 VAC, zero
02	700 VAC, zero
03	mA AC, zero
04	mA DC, zero
05	200 VDC, zero
06	1000 VDC, zero
07	1000 VDC + 20 MΩ
08	20 VDC + 20 MΩ
09	20 VDC + 2000 kΩ
10	2 VDC + 2000 kΩ
11	200 Ω, overrange
12	2 kΩ, overrange
13	20 kΩ, overrange
14	200 kΩ, overrange
15	1000 VDC + X10 T/H + 20 MΩ
16	200 VDC + 200 kΩ
17	200 VDC + 20 kΩ
18	200 VDC + 2 kΩ
19	200 VDC, filter on
20	200 VDC + 2 kΩ, filter off
21	200 VDC, filter off

Digital self-test errors | 25–29

Code	Meaning / test configuration
25	In-Guard microcontroller internal RAM
26	Display RAM
27	In-Guard microcontroller internal program memory
28	External program memory

Code	Meaning / test configuration
29	Calibration memory

Operation, calibration, and interface errors | 30–77

Code	Meaning / test configuration
30	AC functions require the True RMS AC option.
31	mA AC or mA DC selected while REAR inputs are selected.
32	OFFSET selected when reading is unavailable or overrange.
40	Computed calibration constant out of range; prior calibration or hardware may be at fault.
41	Calibration input outside acceptable range; check applied input, prior calibration, and hardware.
42	Calibration memory write error; likely hardware fault.
50	Guard crossing error detected by In-Guard microcontroller.
51	Calibration command requires calibration mode.
52	Command not valid at this time / in current calibration phase.
53	Invalid calibration value in Put command (e.g., negative value in AC calibration).
54	Command not valid during calibration verification.
56	Variable inputs not allowed during A/D calibration; use prompted value.
60	Device-dependent commands not valid during self-tests.
71	Syntax error in device-dependent command string.
72	Guard crossing error detected by Out-Guard microcontroller.
73	Guard crossing error at power-on, or CAL ENABLE switch on at power-on.
77	IEEE-488 interface self-test error.

Running and interpreting self-tests

• Disconnect all test leads, then hold SRQ for approximately 3 seconds to start the diagnostic self-tests.

• Inputs left connected can produce false failures, especially codes 05, 07, 08, 09, and 10.

• With FRONT/REAR set to REAR, tests 03 and 04 are skipped. Without True RMS AC Option -09, tests 01, 02, and 03 are skipped.

• If the A/D converter or precision voltage reference is faulty, many or all analog tests may fail together. If the A/D converter is not calibrated, tests 07, 15, and 19 may fail.

• During display of an analog self-test error, pressing SRQ latches the instrument in that test configuration for troubleshooting. See manual Table 6-18 for expected test-point voltages.

• Codes 01–29 are self-test codes; the other listed codes are operating, calibration, or communication errors, not necessarily failed diagnostic stages.

Source

Fluke 8840A Instruction Manual, Table 2-1 (Error Codes), Section 6-38 (Diagnostic Self-Tests), and Section 6-39 (Self-Test Descriptions). Reference copy: https://www.artisantg.com/info/fluke_8840a_manual.pdf